AGRIVIVE / KHAZAN-GATE

GOA / KHAZAN-INSPIRED / ONE-WAY BUND FLAP

SERIES 06 / 28

A3 / PRELIMINARY / ALL DIMENSIONS mm



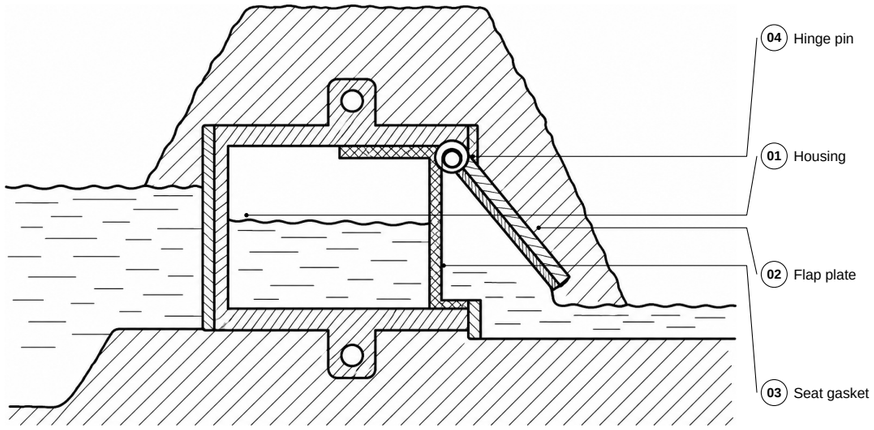
STATE CONTEXT / INDIGENOUS INSPIRATION

Goa's khazan lands use bunds and sluice doors to manage tidal water movement. KHAZAN-GATE explores that one-way principle in a compact trial module. It does not replace a tidal defence.

Practice source: Planning Commission Goa eco-development report, 1982 | Outline: geoBoundaries / schematic

01 INSTALLED SOIL / WATER SECTION

NOMINAL ENVELOPE / mm: 120 W x 100 H x 80 D

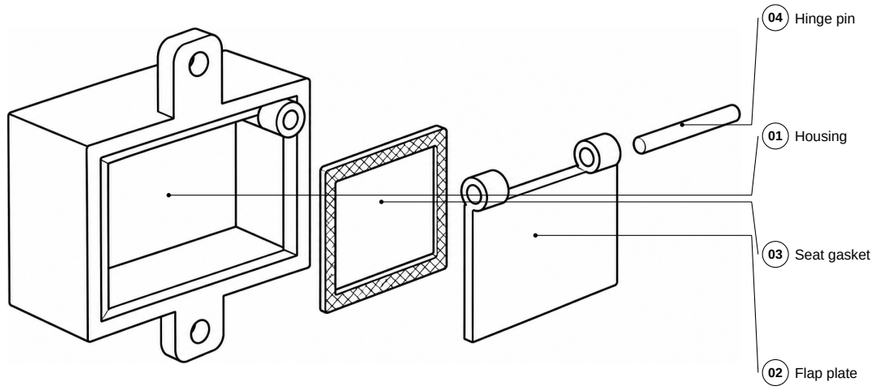


DESIGN / OPERATION NOTES

Install with the hinged flap on the downstream side and hinge above the opening.
Forward differential head can swing the flap outward; reverse head should press it against the seat.
Debris can prevent closure. Keep the hinge, seat and downstream opening accessible.

02 EXPLODED ASSEMBLY

ASSEMBLED ENVELOPE / mm
120 W x 100 H x 80 D

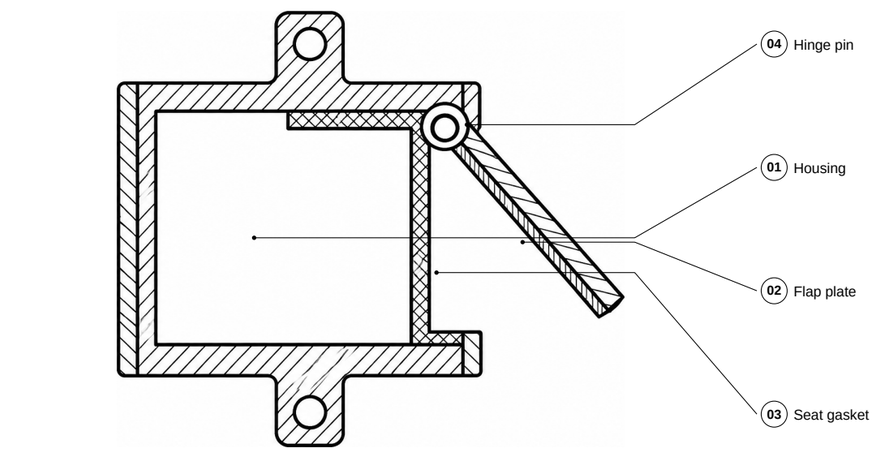


DESIGN / OPERATION NOTES

Components separated for service and assembly.
Connections and mating fits require CAD.

04 INTERNAL SECTION

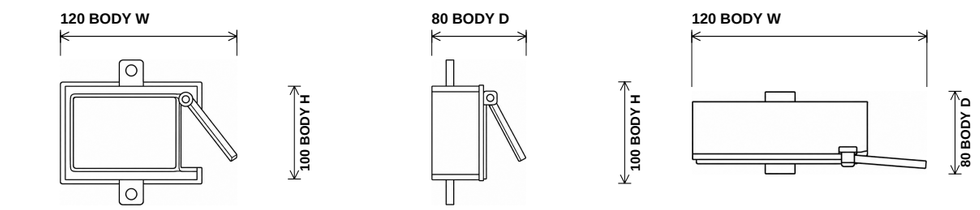
SECTION BASIS / mm
30 anchor projection additional; flap window 80 W x 50 H



DESIGN / OPERATION NOTES

Forward differential head can swing the flap outward; reverse head should press it against the seat.
Debris can prevent closure. Keep the hinge, seat and downstream opening accessible.

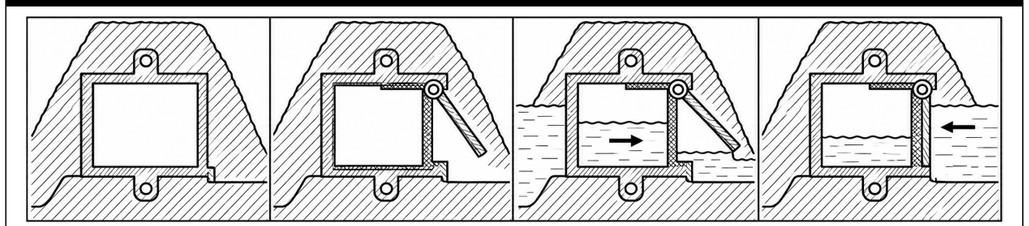
03 ORTHOGRAPHIC STUDIES / mm



UPSTREAM **SIDE** **PLAN**

Front: 120 W x 100 H. Side: 80 flow depth. Clear window 80 W x 50 H.

05 INSTALLATION / OPERATING SEQUENCE



1 FIT HOUSING **2 FIT FLAP** **3 FORWARD FLOW** **4 REVERSE CLOSE**

Prototype fit, access and operation require trials.

DESIGN / INTERFACE SCHEDULE

01 NOMINAL GEOMETRY

External W x H x D	**120 x 100 x 80**
Clear window W x H	**80 x 50**
Anchor projection	**30**
Hinge / gasket detail	**TBD**

02 GEOMETRIC CHECK

CLEAR FLAP WINDOW

4,000.0 mm2

80 x 50
Opening pressure and backflow sealing unverified

03 PARTS / QTY

01 Housing	**1**
02 Flap plate	**1**
03 Seat gasket	**1**
04 Hinge pin	**1**
05 Mounting tabs	**Integral**

Housing/flap: compatible candidate material. Pin and gasket grades TBD.

04 INTERFACES / CHECKS

Test opening head, leakage and hinge friction.

Inspect for debris jamming and gasket deformation.

Validate corrosion and anchorage in site conditions.

PRELIMINARY / VALIDATION PENDING

DESIGN DEVELOPMENT RECORD

01 / GEOMETRY
Nominal body sizes are proposed. Generated views are illustrative; small features and interfaces may vary.

02 / BEFORE FABRICATION
Develop one CAD model, assign wall thicknesses, tolerances and mating fits. Check print orientation and service access.

03 / TRIAL RECORD
Measure dimensions and retained volume or opening area. Record leakage, clogging and soil anchorage under observed conditions.

No discharge, treatment or durability performance is established.

DRAWING / REVISION	TITLE	STATUS	REVISION DATE	FORMAT / SCALE / UNITS	COLLECTION SHEET
AGR-06 / P01	KHAZAN-GATE	PRELIMINARY CONCEPT / CAD & TRIALS PENDING	10 NOV 2026	A3 / NTS / mm	06 / 28